AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Project

Project Title:	Shop Inventory and Billing			
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Course Title:	PROGRAMMING IN PYTHON			
Course Code:	CSC4162		Section:	B
Semester:	Spring	2025-26	Course Teacher:	MD. TANZEEM RAHAT

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Shop Inventory and Billing

Description:
This application is a desktop base application. The main purpose of the app is to manage shop inventory and generate invoice in a simple efficient way. This application has two interfaces. Frist is Inventory Interface and second is Billing Interface. In Inventory Interface we can see the product details and add product, update product details, remove product, search product. In Billing Interface there is add to cart option, remove product from the cart, and checkout option.

Features:
Inventory Features:-

- Add Product
- Update Product
- Delete Product
- View Product List
- Search Product
- Sort Product
- Inventory Summary

Billing Features:-

- Add items to cart
- Remove items from cart
- Clear cart
- Generate invoice

Input Validation and Error Handling:

- No empty product name
- Price must be numeric
- Quantity must be integer
- Cannot exceed available stock

System Workflow:
Inventory Process:-

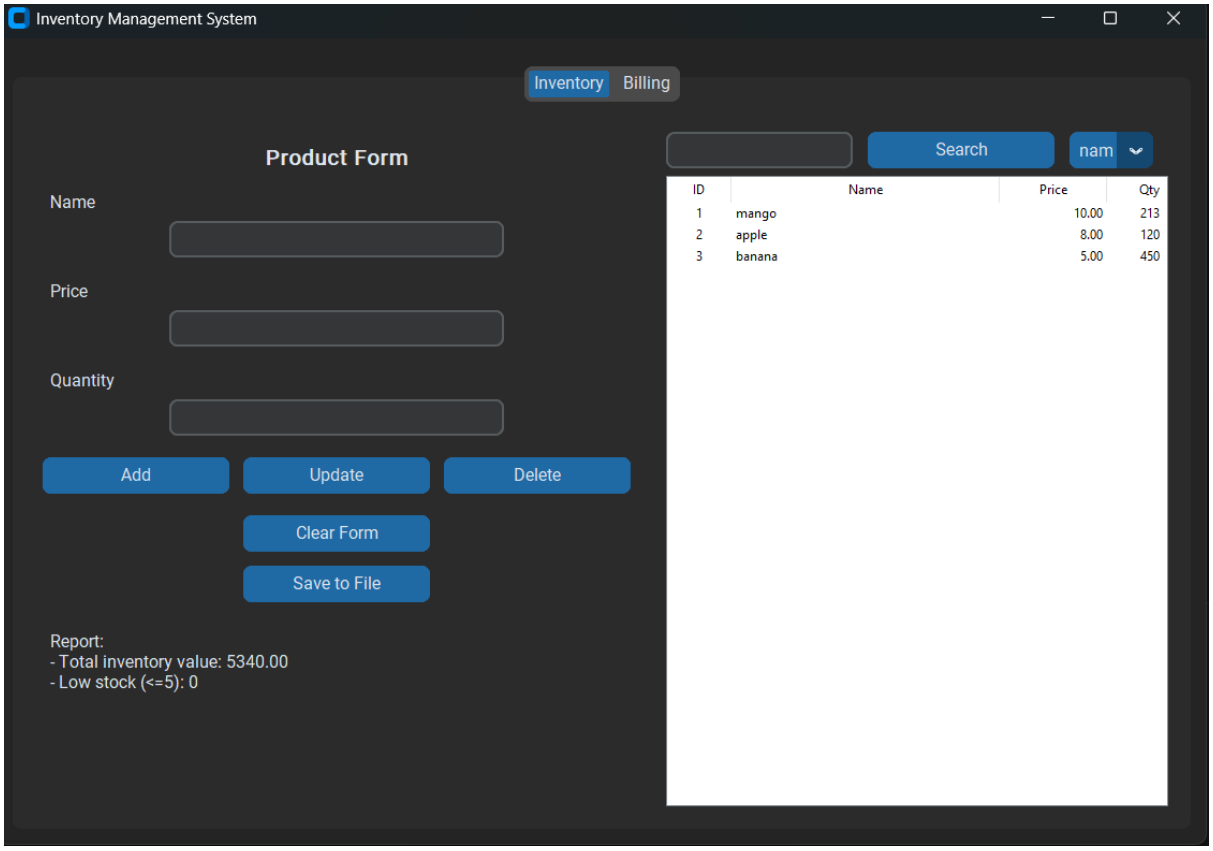
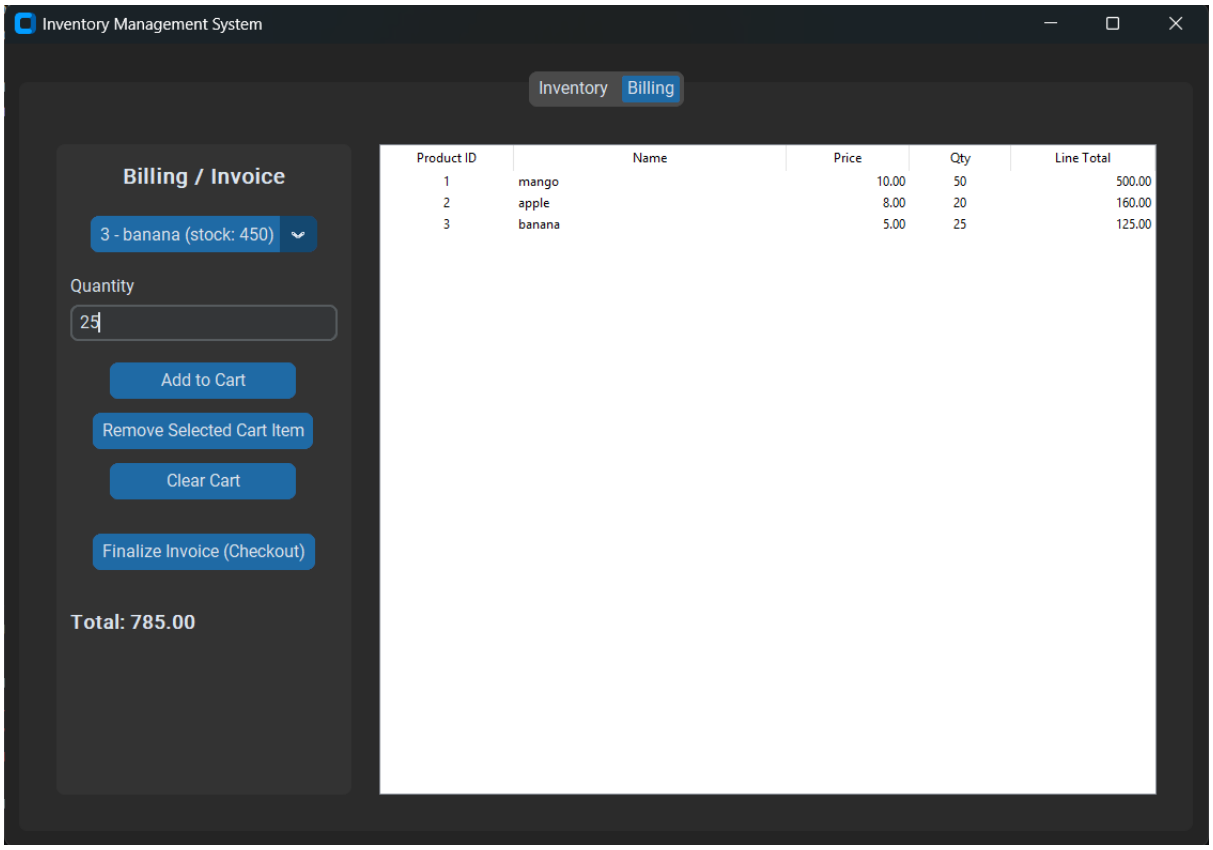
1. Enter product details
2. Add product
3. Edit or delete product
4. Save data

Billing Process:-

1. Select product from select bar
2. Enter the quantity
3. Add to cart or Remove from cart
4. Checkout

What I Have Done:

- GUI with 2 views
- OOP implementation

- JSON data storage
- CRUD operation
- Search and Sorting
- Billing system
- Validation handling

Screenshots: